

**Theorem 8.1** (Multifold sums of powers). *For non-negative integers  $n, m$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n - t + \frac{k}{2} + r}{k + r} + \binom{n - t + \frac{k}{2} + r - 1}{k + r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[ \binom{r - s - t + \frac{k}{2}}{k + r - s} + \binom{r - 1 - s - t + \frac{k}{2}}{k + r - s} \right] \Sigma^s n^0 \right\} \delta^k t^m. \end{aligned}$$

For example,